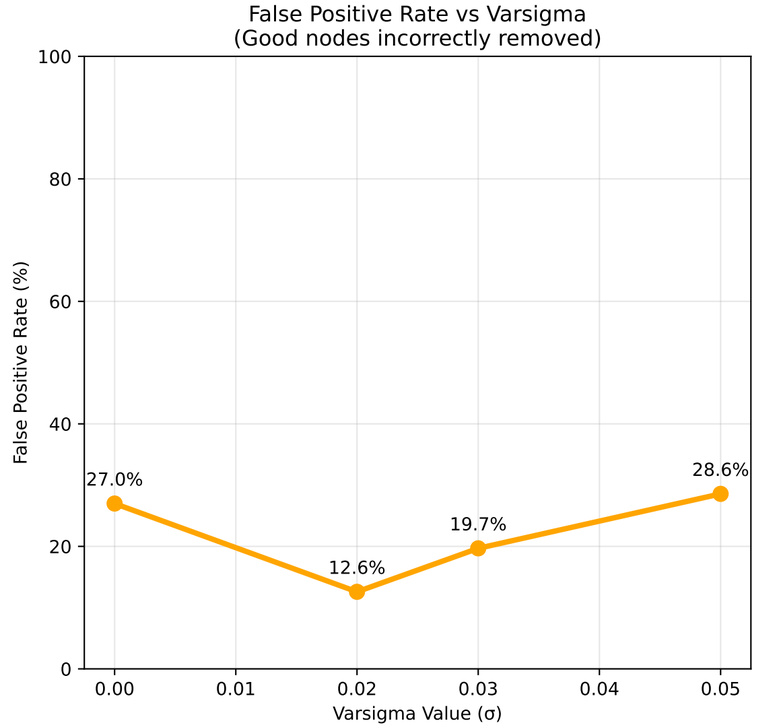
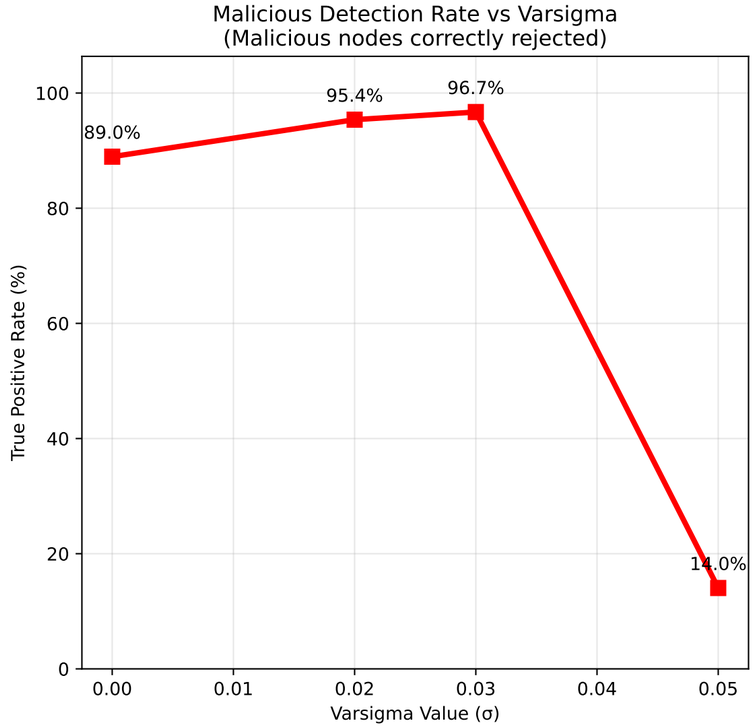
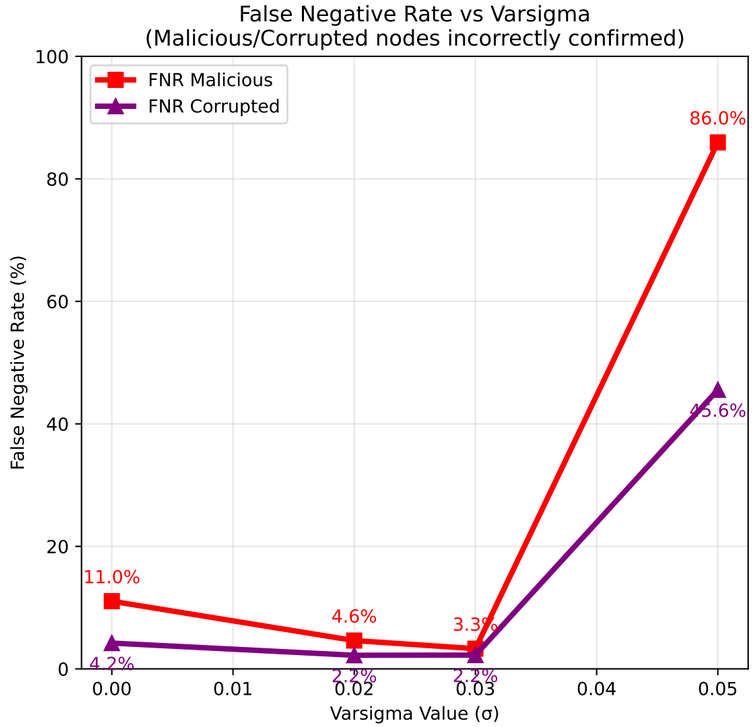
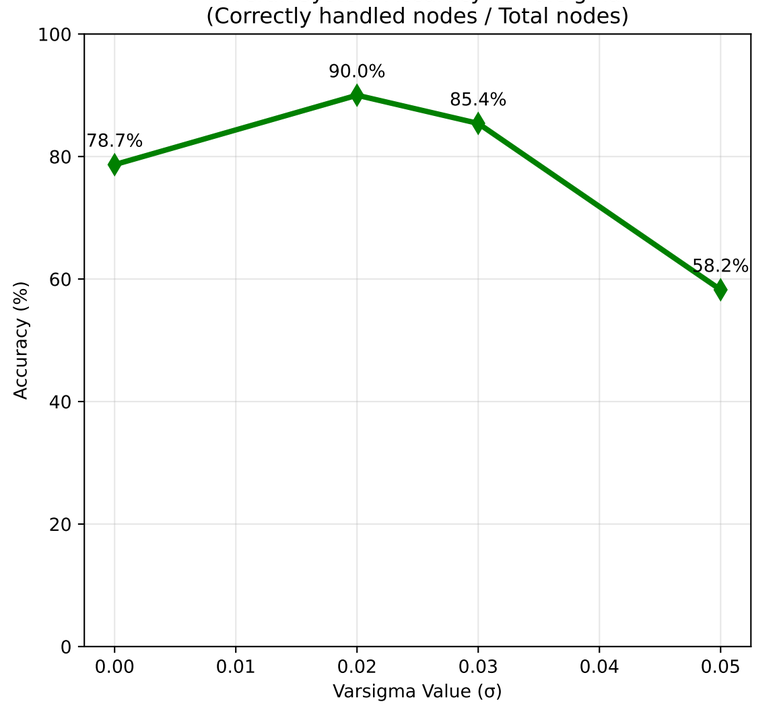
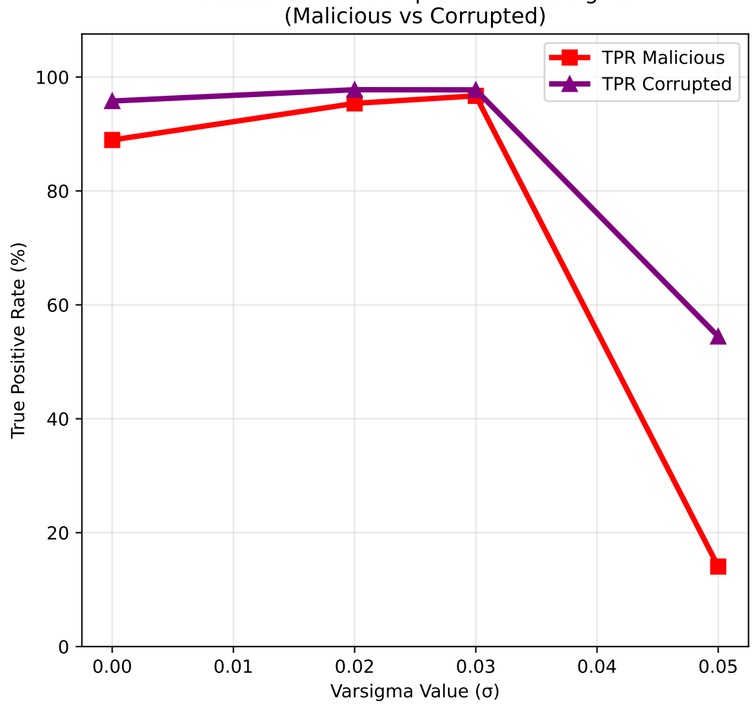
Varsigma Comparison - Cumulative Weight Penalty Factor Analysis

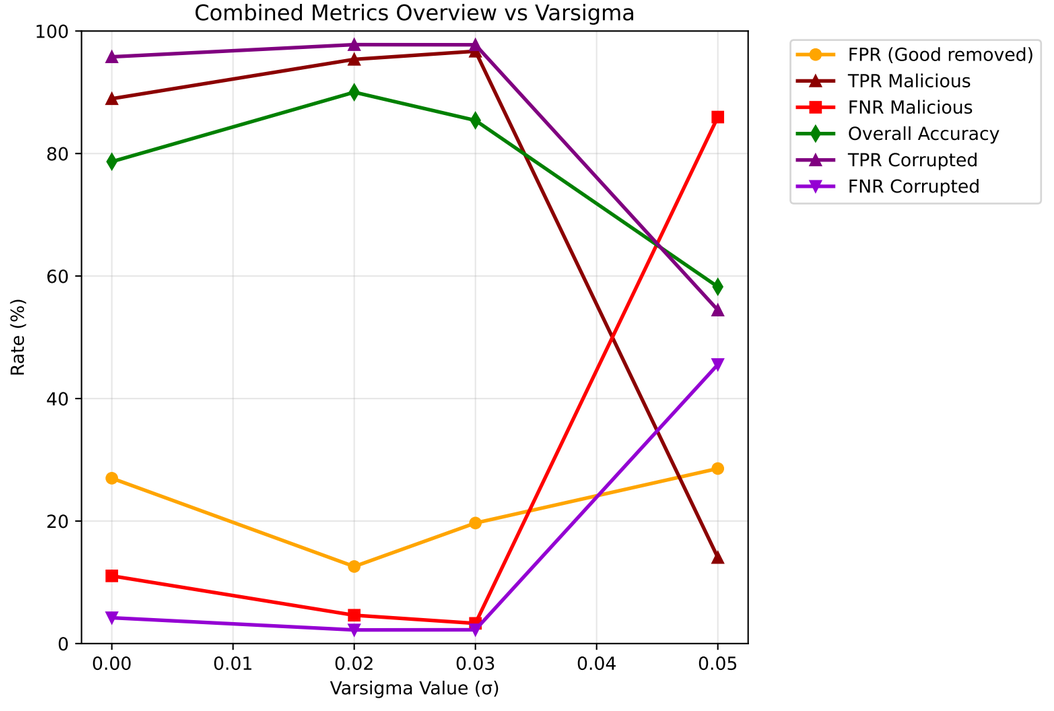












Varsigma Effectiveness Ranking

```
VARSIGMA EFFECTIVENESS RANKING
======================================

1. Varsigma = 0.02
   FPR: 12.6%, TPR: 95.4%, FNR: 4.6%, Acc: 90.0%
   Score: 168.2

2. Varsigma = 0.03
   FPR: 19.7%, TPR: 96.7%, FNR: 3.3%, Acc: 85.4%
   Score: 159.1

3. Varsigma = 0.0
   FPR: 27.0%, TPR: 89.0%, FNR: 11.0%, Acc: 78.7%
   Score: 129.6

4. Varsigma = 0.05
   FPR: 28.6%, TPR: 14.0%, FNR: 86.0%, Acc: 58.2%
   Score: -42.2


VARSIGMA PARAMETER:
The varsigma parameter controls the cumulative weight penalty
factor in the TrustInformedTipSelector algorithm.
Higher varsigma values apply stronger penalties to nodes
with high cumulative weights.

VARSIGMA VALUES TESTED:
Values: [0.0, 0.02, 0.03, 0.05]
Configuration: Testing cumulative weight penalty effects
```

Summary Statistics

```
VARSIGMA COMPARISON ANALYSIS - SUMMARY
======================================

Generated: 2025-08-31 00:01:05
Results Path: FL/results_varsigma

CONFIGURATION ANALYZED:
• Cumulative Weight Penalty Factor Testing
• Varsigma values tested: [0.0, 0.02, 0.03, 0.05]

DETAILED RESULTS:
\n• Varsigma 0.0:\n  FPR=27.0%, TPR=89.0%, FNR=11.0%, Acc=78.7%\n\n• Varsigma 0.02:\n  FPR=12.6%, TPR=95.4%, FNR=4.6%, Acc=90.0%\n\n• Varsigma 0.03:\n  FPR=19.7%, TPR=96.7%, FNR=3.3%, Acc=85.4%\n\n• Varsigma 0.05:\n  FPR=28.6%, TPR=14.0%, FNR=86.0%, Acc=58.2%\n

METRICS EXPLANATION:
• FPR (False Positive Rate): Good nodes incorrectly removed / Total good nodes sent
• TPR (True Positive Rate): Malicious nodes correctly detected / Total malicious nodes sent
• FNR (False Negative Rate): Malicious nodes incorrectly confirmed / Total malicious nodes sent
• Accuracy: Overall system accuracy (correctly handled nodes / total nodes)

CALCULATION METHOD:
• Each round selects K-1 = 9 nodes randomly from K = 10 peers
• Uses actual worker selection data from 3000_workers_selected.csv
• Sums Good_Nodes_Removed_This_Round and Malicious_Nodes_Confirmed_This_Round across ALL rounds
• Counts actual occurrences of each peer type in selected workers

OPTIMAL VALUES:
• FPR: 0% (no good nodes removed)
• TPR: 100% (all malicious nodes detected)
• FNR: 0% (no malicious nodes confirmed)
• Accuracy: 100% (perfect classification)

VARSIGMA PARAMETER:
The varsigma parameter controls the cumulative weight penalty factor
in the TrustInformedTipSelector algorithm. It determines how much penalty
is applied to nodes with high cumulative weights:
• Varsigma = 0.00: No penalty for cumulative weight
• Varsigma = 0.02: Light penalty (default value)
• Varsigma = 0.03: Moderate penalty
• Varsigma = 0.05: Strong penalty
• Higher varsigma values may improve malicious detection but can penalize active good nodes
```